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Research Paper

Thermal properties optimization of microencapsulated a renewable and non-toxic phase change material with a polystyrene shell for thermal energy storage systems

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HIGHLIGHTS

- Microencapsulated of lauric acid with a polystyrene shell was prepared.
- A renewable and non-toxic phase change materials has been used for the process.
- Emulsion polymerization has been applied as a microencapsulated technique.
- Response surface method was applied to the statistical design.
- The microencapsulated PCM had a good potential for energy storage system.

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ABSTRACT

Thermal energy storage (TES) plays an important role in the development of an efficient solar energy system by storing the solar energy when available during the daytime and use it at night when required. The main component of a TES system is encapsulated phase change materials in macro, micro and nano sacles. Microencapsulated lauric acid (LA) as a renewable (obtained from vegetable oils) and non-toxic Phase Change Material (PCM) with a polystyrene shell was prepared using an emulsion polymerization technique. The individual and interactive effects of operating variables including lauric acid to styrene mass ratio, sodium dodecyl sulfate (SDS) to styrene (St) mass ratio, stirring rate and temperature on the microencapsulation ratio (ME.R) were investigated. Response surface method was applied to the statistical design, analysis of experiments and process optimization. Analysis of Variance (ANOVA) showed that the interaction between the stirring rate and temperature had non-significant effects on ME.R (%). The maximum achieved value of ME.R was 92.3% in the process optimization. It was enhanced compared with microencapsulation ratio of lauric acid in previous studies. By using the optimal values, LA/St mass ratio of 0.42, emulsifier (SDS) to styrene mass ratio of 0.01, stirring rate of 1076 rpm and the temperature of 55 °C, ME.R of 91.64% was obtained. Thermal properties, morphology and thermal stability of the optimal microcapsules were studied using DSC thermograms, Scanning Electron Microscopy (SEM) and thermogravimetry analysis (TGA), respectively. The results showed that microencapsulated renewable PCM with a melting point of 43.77 °C and latent heat of 167.26 kJ/kg has a good potential for utilizing renewable solar energy.

1. Introduction

Nowadays, renewable energy as a great source of energy is important due to the limited fossil fuels and the increase of energy demand [1]. Thermal Energy Storage (TES) which provides the matching of energy supply and demand has been developed during the last twenty years from small-scale laboratory setups to an important technology for the future which affect the economy and the environment and it is a key technology that can also increase the potential of utilizing renewable energy sources such as ambient cold air or solar energy. There is always an imbalance between production and consumption of energy in solar energy systems due to intermittent and unpredictable nature of solar radiation. Phase change materials (PCMs) with their special features reduce this imbalance between demand and consumption.

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Therefore, they improve performance of energy storage systems and reduce costs due to the wasted energy. PCMs are divided into two groups: inorganic and organic PCMs. Liu and Chang [2] showed that inorganic PCMs with and without additives are not suitable for energy storage applications due to congruent melting, high subcooling and instability during thermal cycles. Organic PCMs are divided into two categories: paraffins and nonparaffins. Fatty acids, as nonparaffin renewable materials, are the main components of triglycerides which are the molecules of vegetable oils. Each molecule of triglyceride is composed of three molecules of fatty acids which linked to the one molecule of glyceride or glycerin. They are a good choice due to stable physical and chemical properties, non-toxicity, low cost, non-flammability, low volume changes, no phase separation, high latent heat, no subcooling and compatibility with all carriers except plastic at high temperatures [3,4]. A useful method for improving thermal conductivity of PCMs and preventing leakage during the melting process is microencapsulation, which is a main obstacle to the industrial applications [5]. The final goal in the microencapsulation of PCMs is not only easy and safe use of them, but also is reducing reactivity and improving thermal properties by increasing the heat transfer surface. Micro and nano-encapsulated PCMs have numerous benefits such as leakage prevention of the melted PCM, control of the volume change during phase change and protection of destructive environmental interactions [6]. These properties of microencapsulation can make PCMs suitable for thermal-regulating solar energy [7], building materials [5,8] and textiles [9]. Several techniques have been used for micro or nano-encapsulation of PCMs, such as interfacial polymerization [10], complex coacervation [11], in situ polymerization [7], spray dryer [12], sol-gel [13], suspension polymerization [14] and emulsion polymerization [15].

An extensive review of the microencapsulation methods has been done by Jamekhorshid et al. [16]. Microencapsulated PCM with polymeric wall materials such as melamine formaldehyde resin, urea formaldehyde resin, polyurethane, and polystyrene or poly methyl methacrylate was obtained by in situ polymerization or interfacial polymerization in an emulsion system [17,18]. Most of the produced microcapsules have an appropriate core-shell structure. The polymeric wall materials play a key role in improving the structural stability and permeability of PCM [19] but disadvantages such as flammability and low thermal conductivity of the microcapsules due to the polymeric shell should also be considered [20,21].

A review of the previous research reveals that there is a lack of studies regarding the micro and nano-encapsulation of fatty acids as renewable phase change materials. The functional group of fatty acids differentiates them from paraffins in micro and nano-encapsulation process. In addition, it is necessary to produce microcapsules with the harder shell to prevent leakage of core material during unlimited number of thermal cycles. Tahani et al. [22] investigated the synthesis of microencapsulated palmitic acid with aluminum hydroxide oxide ($Al_2O_3{\cdot}xH_2O$) as a shell using the sol-gel method. Aluminum isoperoxide is used as a precursor for the shell material. The effects of four different mass ratios of core to shell material on the thermal properties of the obtained microcapsules were studied. The average diameter of the microcapsules with the mass ratios of 50:50 to 20:80 was about 1.689 to 3.730 μm, respectively. The results showed the microencapsulated palmitic acid had better thermal performance due to improving the specific heat and thermal stability. Also, the mass ratio of core to shell was found to play an important role in the morphology, microencapsulation ratio and stability of micro particles. The maximum amount of improvement in thermal conductivity in comparison with the pure PCM was 87.94% and microencapsulation ratio was 66.86%. Tahani et al. [23] in another study produced nano capsules containing stearic acid with the titania shell by the sol-gel method. They studied the effect of the mass ratio of core to shell on the morphology, thermal performance and thermal conductivity of the prepared nanocapsules. Stearic acid was encapsulated with the maximum and minimum diameter of 946.6 and 583.4 nm, respectively. Encapsulation ratios were obtained to be between 30.36–64.76%. The obtained nanocapsules with the high mass ratio of core to shell, showed better phase change properties and more encapsulation ratio, despite the low thermal conductivity. Also, these nanocapsules had good thermal and chemical stability after 2500 melting/solidification cycle.

Cao et al. [24,25] prepared microencapsulated paraffin and palmitic acid with titanium dioxide as a shell by the sol-gel method. Microencapsulated paraffin and palmitic acid were obtained to be about the diameters of 50 and 200–400 nm, respectively. Encapsulation ratios of paraffin and palmitic acid were determined to be 87 and 30% respectively. Alkan and Sari [26] made the stabilized shape of fatty acids by encapsulation. In this study, stearic acid (SA), palmitic acid (PA), miristic acid (MA) and lauric acid (LA) were microencapsulated with poly(methyl methacrylate) using the solution casting method. Samples were prepared with different mass fractions of fatty acids to PMMA (50, 60, 70, 80 and 90% w/w). The best encapsulation ratio was obtained in the mass fraction of 80:20.

Konuklu et al. [16] studied microencapsulated of caprylic acid with different shells such as urea formaldehyde resin, melamine formaldehyde resin and urea melamine formaldehyde resin. The results showed that the best shell material and emulsifier were urea formaldehyde resin and Teen40, respectively. Melting and solidification latent heats of microcapsules with urea formaldehyde shell were determined to be 93.9 and 106.1 kJ/g; also, the average diameter of particles was about 200 nm-1.5 μm.

Emulsion temperature in the microencapsulation process is an important parameter. Microcapsules were made with urea formaldehyde resin in the presence of the emulsifier TEEN40 at 50, 60, 70, 80 and 90 °C, respectively. The best emulsion temperature was between 50 to 70 °C. Also, the emulsification process was investigated in three stirring times of 60, 120 and 160 min, and the appropriate time was obtained to be 120 min. The results showed that particles with the melamine shell were more spherical while the type of emulsifier played an important role in the structure of the particles. Ozonur et al. [27] carried out the microencapsulation of the natural coconut fatty acid with different shells such as Arabic gum, gelatin powder, melamine-formaldehyde, urea and naphthol by coacervation. Resins like urea-formaldehyde, melamine-formaldehyde and β naphthol-formaldehyde were made by simple coacervation and gelatin- Arabic gum was created by complex coacervation. The best wall material was gelatin-Arabic gum. The coconut fatty acid did not emulsify in urea, melamine and beta naphthol solutions; therefore, the polymerization reaction did not occur. Produced microcapsules were able to endure heating and cooling operations at a thermal energy storage system with a temperature range of 22–34 °C after 50 thermal cycles. Preparation of microencapsulated caprylic acid particles with urea formaldehyde (UF), melamine formaldehyde (MF) and melamine urea formaldehyde (MUF) resin was also done by Konuklu et al. [28]. The size of microparticles with PUF, PMF and PMUF was 0.66, 0.32 and 0.29 μm, respectively. Microcapsules with melamine formaldehyde shell had a smaller core and larger shell sizes in comparison with the core and shell of microcapsules with urea formaldehyde shell. The thermal resistance of microcapsules containing poly melamine formaldehyde and poly melamine urea formaldehyde was found to be more than that of microcapsules containing poly urea formaldehyde. Microcapsules containing PMUF with surfactants Teen40 and Teen80 had an irregular structure; on the contrary, in the presence of a combination of both surfactants, they had spherical structures without fractures. The results showed that although preparation of microcapsules was strongly influenced by the surfactant, the thermal stability of microcapsules was not affected by the type of the surfactants. Microencapsulation of Lauric acid with melamine formaldehyde resin was performed by Bao et al. [29]. Microcapsules were synthesized by in situ polymerization, with the mass ratio of core being shell 1:1. The encapsulation ratio was obtained to be 46.2%. In addition, the obtained microcapsules were mixed with acrylic latex and used to make a smart cover. The encapsulation ratio was obtained to be 46.2%, which was the weight of encapsulated lauric acid. After 70 thermal cycles, a distinct change did not occur in the melting temperature, but the latent heat showed an obvious reduction.

Literature review clearly shows that the majority of research has been carried out on the microencapsulation of different types of paraffin materials within the polymer shell. However, studies regarding fatty acids in this field are limited. Emulsion polymerization method is simple, feasible, eco-friendly and able to obtain repeatable results among various polymerization methods. This method is more promising due to the ease and the relatively high microencapsulation ratio [30,31]. Thus, in the present study, Due to the high latent heat and low volume changes of LA in compared with paraffin, the microencapsulation of LA was considered using emulsion polymerization method in our previous study for the first time [32]. The aim of this study was to develop a facile process for the microencapsulation of lauric acid, as a renewable and non-toxic phase change material which is existed in most of the edible and non-edible vegetable oils, by the emulsion polymerization process. The optimal conditions of four factors including lauric acid to styrene mass ratio, emulsifier to styrene mass ratio, temperature and stirring rate were studied to obtain the maximum amount of microencapsulation ratio by the response surface method (RSM). Also, the individual and interactive effects of operating variables on the microencapsulation ratio were investigated. Thermal properties, morphology and thermal stability of the optimal microcapsules were studied using DSC thermograms, Scanning Electron Microscopy (SEM) and thermogravimetry analysis (TGA), respectively.

2. Materials and methods

2.1. Materials

Styrene (purity > 99 wt%, Merck Chemical Co.) was used as a monomer for forming the microcapsules shell. Styrene was washed three times with 5 wt% NaOH solution to remove the inhibitor. Then it was sprayed for four times with distilled water to ensure that sodium hydroxide is removed completely. Luaric acid (LA) with melting point of 45 °C, heat of fusion of 182.52 kJ/kg, thermal conductivity of 0.147 W/m K at 50 °C and density of 870 kg/m^3 at 50 °C was used as a renewable and non-toxic PCM which was of a reagent grade ($C_{12}H_{24}O_2$, Merck Chemical Co.) for an oily core. Potassium persulfate of synthesis grade (purity > 97 wt%, Merck Chemical Co.) was used as an initiator. Sodium dodecyl sulfate ($C_{12}H_{25}OSO_3Na$, SDS) was of reagent grade (purity > 96 wt%, Merck Chemical Co.) and was used as an oil–water emulsifier. Methanol (purity > 97 wt%, Merck Chemical Co.) and deionized water were used to pour the samples.

2.2. Microencapsulation procedure

Emulsion polymerization was carried out in a 2-L three neck glass reactor which has been applied in our previous work [32] that was equipped with the digital control of the stirring rate, oil thermostat bath, and a reflux condenser under nitrogen gas. The process involved the following stages: (i) SDS was dissolved in 80 ml deionized water as the aqueous phase in the reactor at 30 °C under the mild agitation. (ii) Lauric acid was added to the reactor in the melting point of PCM. Then, stirring rate and temperature was raised to the desired amount (iii) in the third step, after 15 min potassium persulfate as an initiator was dissolved in deionized water, and poured to the reactor mixture. (iv) Styrene was poured into the reactor drop by drop. The polymerization process was carried out at atmosphere pressure for 24 h under the nitrogen flow. The obtained PCM microcapsules were washed with methanol and deionized water and filtered to remove the impurities. The purified microcapsules were dried at the ambient temperature for 24 h. DSC thermograms of each sample was analyzed for calculating the microencapsulation ratio and Response Surface Methodology (RSM) was employed to determine the optimal conditions for high microencapsulation ratio of lauric acid with the poly(styrene) shell.

2.3. Experimental design and data analysis

Response Surface Method (RSM) includes optimization methods. This method can simplify complex problems to simple problems by collection of statistical techniques. RSM determines the sensitivity of response versus each independent factor. One of the most common types of experimental design is the central composite design (CCD). In this design, four factors: lauric acid to styrene mass ratio (LA/St), emulsifier to styrene mass ratio (SDS/St), stirring rate and temperature (T) were selected as independent variables. The explanation for selecting these independent variables has been represented in our previous work [32]. The experiments were designed randomly by RSM based on the CCD, which included eight axial points, sixteen cube points, and seven center points in cube. In the full factorial method each of independent variables has 3 levels; a center point and two factorial points with the distance of ±1 from center point. Four independent variables and their coded and actual levels are shown in Table 1. Z_i and X_i are coded and actual independent variables, respectively.

Microencapsulation ratio (ME.R) of micro particles was selected as response variable. It can be defined as the ratio of the latent heat of microencapsulated lauric acid to the latent heat of the pure lauric acid. The lauric acid microencapsulation ratio was calculated by using the following equation [29,32]:

$$ME.R = \frac{\Delta H_{MEPCM}}{\Delta H_{LA}} \times 100 \quad (1)$$

where ΔH_{MEPCM} and ΔH_{LA} refer to the latent heats of microencapsulated lauric acid and pure lauric acid, respectively. The latent heats were calculated from DSC thermograms. The Microencapsulation ratio was fitted by a quadratic polynomial regression model. The model proposed for the response of *ME.R* fitted Eq. (2), as follows:

$$ME.R = \beta_0 + \sum_{i=1}^{3} \beta_i Z_i + \sum_{i=1}^{3} \beta_{ii} Z_i^2 + \sum_{i=1}^{2} \sum_{j=i+1}^{3} \beta_{ij} Z_i Z_j \quad (2)$$

where *ME.R* is response variable (%), and β_0, β_i, β_{ii} and, β_{ij} are constant coefficients of intercept, linear, quadratic and interaction terms, respectively. Z_i is coded independent variables.

2.4. Sample characterization

The phase change melting point and latent heat of samples was determined using differential scanning calorimeter (DSC), BAHR Thermoanalyse GmbH (Germany) with the heating rate of 5 °C/min and in a temperature range of 25–70 °C under nitrogen atmosphere.

Table 1
Coded and actual levels of independent variables used in the experimental design.

LA/Styrene (mass ratio)		SDS/Styrene (mass ratio)		Stirring Rate (rpm)		Temperature (°C)	
Coded (Z_1)	Actual (X_1)	Coded (Z_2)	Actual (X_2)	Coded (Z_3)	Actual (X_3)	Coded (Z_4)	Actual (X_4)
−1	0.2	−1	0.01	−1	500	−1	50
0	0.4	0	0.035	0	1000	0	55
1	0.6	1	0.06	1	1500	1	60

The thermal stability test was determined using thermogravimetry analysis (TGA) under a constant stream of argon at a flow rate of 20 ml/min, heated from 25 to 600 °C with the heating rate of 10 °C/min. The morphology and structure of the micro particles was investigated using a KYKY-EM3200 Scanning Electron Microscope system (SEM). The average diameter of particles can be analyzed with ImageJ software. The Fourier transform infrared (FT-IR) was used for the identification of both lauric acid and poly (styrene) in the microcapsule samples. FT-IR spectra of the samples were obtained using a Tensor 27 FT-IR spectrophotometer (Bruker Optik GmbH, Madrid, Spain) with dry KBr pellets prepared using a manual hydraulic press. FT-IR spectra in absorbance mode were recorded (200 scans with 4 cm^{-1} resolution) in the range of 400–4300 cm^{-1}.

3. Results and discussion

3.1. Model equation

Thirty-one polymerization experiments were designed using CCD methodology. The experimental error was determined by 7 replicates at the center points. The experimental conditions and their responses are shown in Table 2. The results inserted to Minitab 16 software and the models were fitted to microencapsulation ratio (ME.R (%)).

The analysis of variance (ANOVA) of the models was shown in Table 3. The p-value and the F-value in the Analysis of Variance (ANOVA) is useful tool to check significant of model. In this study, coefficients with $p \leq 0.001$, $0.001 \leq p \leq 0.05$ and $0.05 \leq p$ were considered highly significant, significant and non-significant respectively. It illustrates that among four models in Table 3, 2FS and quadratic models are significant in 95% confidence level ($p \leqslant 0.05$).

Quadratic model was selected for prediction of ME.R response because of high R-squared and F-value compared to others.

Final quadratic model for the prediction of the response variables is given by the Eq. (3) in terms of coded variables (Z_i) regardless of estimated coefficients. Where Z_1 is lauric acid to styrene (LA/St) mass ratio Z_2 is SDS to styrene (SDS/St) mass ratio, Z_3 is stirring rate and Z_4 is temperature.

$$\begin{aligned} ME.R(\%) = {} & 89.683 + 3.993Z_1 - 4.561Z_2 + 2.217Z_3 - 1.531Z_4 \\ & - 3.212Z_1^2 - 1.942Z_2^2 - 15.742Z_3^2 - 2.742Z_4^2 \\ & - 7.298Z_1Z_2 - 3.033Z_1Z_3 + 5.043Z_1Z_4 \\ & - 1.899Z_2Z_3 - 5.678Z_2Z_4 - 1.411Z_3Z_4 \end{aligned} \tag{3}$$

Table 2
Experimental conditions of polymerization tests including experimental and model results for the response.

Run.	LA/St	SDS/St	Stirring rate (rpm)	Temperature (°C)	ME.R (%)	
	Z_1	Z_2	Z_3	Z_4	Experimental	RSM Model
1	1	1	1	1	51.540	51.886
2	1	1	1	−1	60.940	59.040
3	−1	0	0	0	85.210	82.478
4	0	0	0	0	90.560	89.683
5	0	1	0	0	83.910	83.180
6	0	0	−1	0	73.540	71.725
7	1	−1	1	−1	73.470	75.201
8	−1	1	−1	1	70.160	72.280
9	0	0	0	0	90.040	89.683
10	1	1	−1	1	58.020	60.139
11	0	0	0	0	85.620	89.683
12	0	0	0	−1	90.540	88.472
13	1	−1	−1	−1	68.810	70.214
14	−1	−1	−1	−1	52.040	51.651
15	−1	−1	1	1	62.820	64.157
16	−1	1	−1	1	52.370	50.596
17	1	0	0	0	88.480	90.465
18	1	−1	−1	1	93.830	91.415
19	0	0	0	0	89.810	89.683
20	−1	1	1	−1	79.430	81.802
21	0	0	0	0	87.670	89.683
22	1	−1	1	1	92.650	90.760
23	0	0	0	0	90.610	89.683
24	0	0	0	0	91.230	89.683
25	0	−1	0	0	92.320	92.302
26	−1	−1	−1	1	50.550	52.680
27	0	0	0	1	84.090	85.410
28	1	1	−1	−1	63.030	61.650
29	−1	−1	1	−1	70.660	68.770
30	−1	1	1	1	55.650	54.476
31	0	0	1	0	75.090	76.158

Table 3
The analysis of variance (ANOVA) of the models.

Response	Model	Sum of square	DOF	Mean square	R-squared	F-value	P-value
ME.R (%)	Linear	792.15	4	198.038	12.11%	0.90	0.481
	2FS[a]	4431.30	8	553.912	67.76%	5.78	0.000
	2FI[b]	2803.90	10	280.390	42.88%	1.50	0.211
	Quadratic	6443.04	14	460.217	98.53%	76.46	0.000

[a] Two-factor square.
[b] Two-factor interaction.

Table 4
Estimated regression coefficients for ME.R (%) in quadratic polynomial model.

Term	Coefficient	Standard error	P-value
Constant	89.683	0.7278	0.000
Z_1	3.993	0.5783	0.000
Z_2	−4.561	0.5783	0.000
Z_3	2.217	0.5783	0.001
Z_4	−1.531	0.5783	0.018
Z_1^2	−3.212	1.5229	0.051
Z_2^2	−1.942	1.5229	0.220
Z_3^2	−15.742	1.5229	0.000
Z_4^2	−2.742	1.5229	0.091
$Z_1 \times Z_2$	−7.298	0.6133	0.000
$Z_1 \times Z_3$	−3.033	0.6133	0.000
$Z_1 \times Z_4$	5.043	0.6133	0.000
$Z_2 \times Z_3$	−1.899	0.6133	0.007
$Z_2 \times Z_4$	−5.678	0.6133	0.000
$Z_3 \times Z_4$	−1.411	0.6133	0.035

Estimated regression coefficients for ME.R (%) in quadratic polynomial model are shown in Table 4. The P-values of independent variables of Eq. (3) in Table 4 show that among these variables, only the quadric term of SDS/St mass ratio (Z_2^2) and temperature (Z_4^2) have a non-significant effect on the microencapsulation ratio (ME.R). The rest of the terms have positive and negative significant effects on ME.R (%).

Predicted values of the quadratic model versus experimental values of the ME.R (%) were shown in Fig. 1. The obtained values of R^2,$R^2(Adjusted)$ and $R^2(predicted)$ were shown in Fig. 1. These values are 98.53%, 97.24% and 91.82%, respectively. The results show the predicted data from quadratic model are in agreement with the experimental ones.

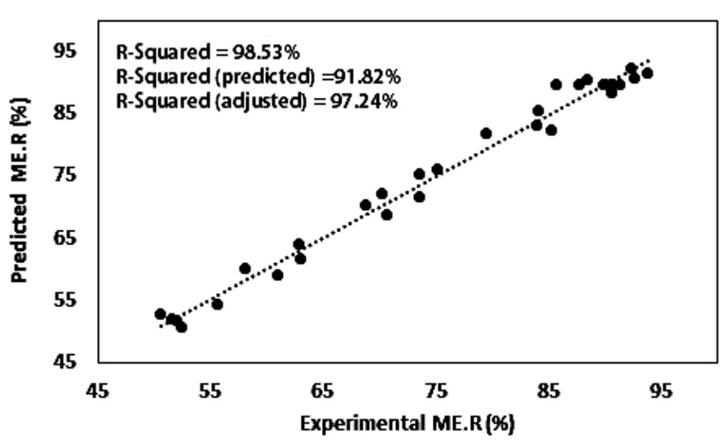
3.2. Interaction between process parameter

The interaction plot of independent variables on ME.R (%) is presented in Fig. 2. When the lines in interaction plot are parallel, the independent variables do not have interaction effect on each other. It can be seen in Fig. 2 that almost all lines are intersecting to each other. The square in the bellow right side of Fig. 2 shows that stirring rate has significant interaction with temperature on ME.R (%). The P-values of interactions values in Table 4 confirm that the interaction between stirring rate and temperature has significant effects on ME.R (%).

Fig. 3(a) shows the effect of stirring rate and SDS/St mass ratio on ME.R (%). The microencapsulation ratio is increased from 71.725 to 89.683% by increasing the stirring rate from 500 (−1) to 1000 (0). It is decreased to 76.158% when stirring rate increased to 1500 (1). Therefore, stirring rate obtains an optimum amount as can be seen from Fig. 4(b). Dispersion capability of lauric acid in aqueous solution is better than paraffin materials due to their carboxyl functional groups. As a matter of fact, required amount of SDS to dispersion of fatty acids in aqueous phase will be less. The excessive amount of SDS results decreasing particle size and enhancing the specific surface area, in presence of certain value of styrene, specified amount of lauric acid can be coated with poly (styrene) and ME.R reduces, consequently (Fig. 3(b)).

3.3. Optimization

The aim of the response surface optimization method is detecting desirable values of independent variables where the response could be maximum or minimum. In this study, RSM by Minitab 16 was used to maximize the microencapsulation ratio (ME.R (%)) at the optimal conditions. The RSM results showed the maximum $E.R$ = 92.3% was achieved in values of the independent variables; LA/St mass ratio = 0.42, SDS/St mass ratio = 0.01, stirring rate = 1076 rpm and temperature = 55 °C. The optimum range of variables except the SDS/St mass ratio was similar the optimal condition of microencapsulated paraffin with poly (styrene) in our previous work [32]. The optimal SDS/St mass ratio in the microencapsulation of lauric acid was less than that of the microencapsulation of paraffin. It could be due to existence of carboxyl functional group in fatty acid structure. The carboxyl group could



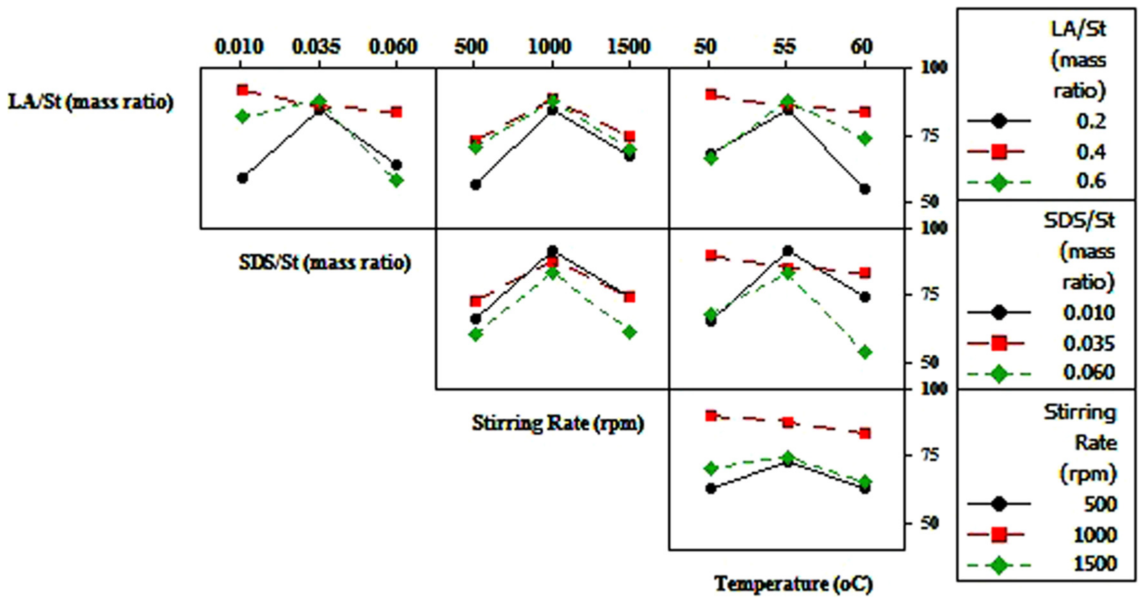


Fig. 2. Interaction plot of independent variables on ME.R (%).

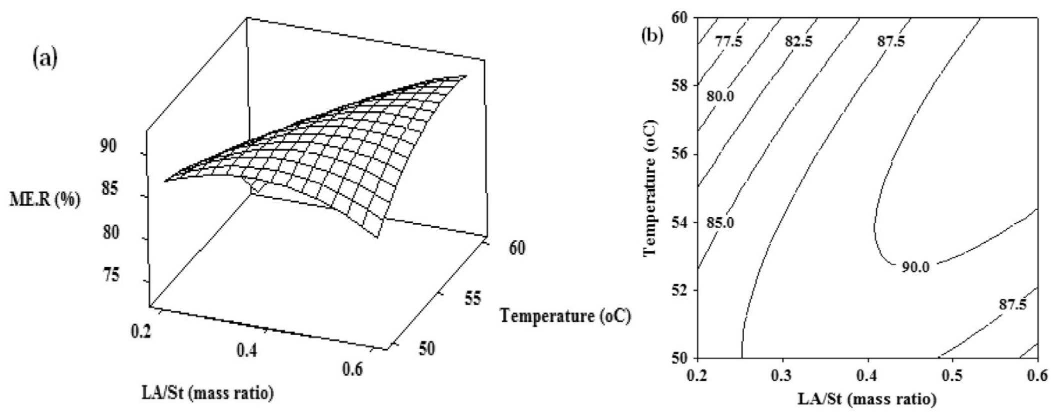


Fig. 3. The response surface (a) and contour (b) plots for ME.R (%) as a function of uncoded values of LA/St mass ratio and temperature at the constant SDS/St mass ratio of 0.035 and stirring rate of 1000 rpm.

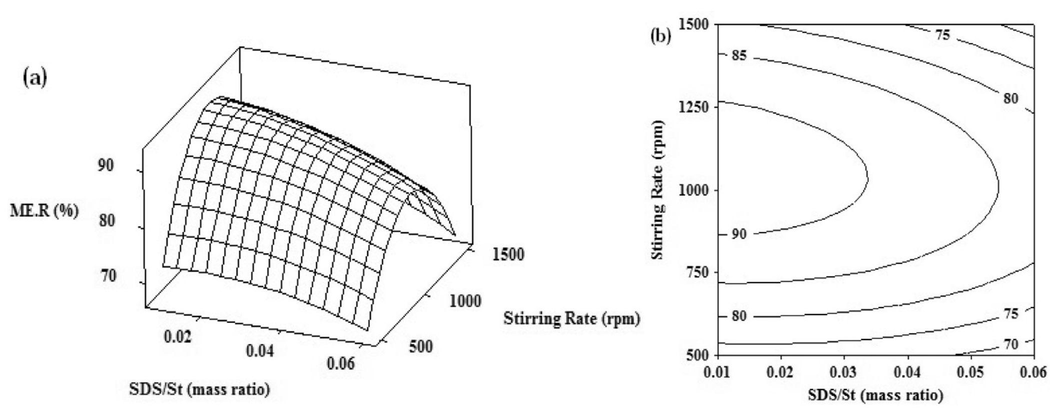


Fig. 4. The response surface (a) and contour (b) plots for ME.R (%) as a function of uncoded values of SDS/St mass ratio and Stirring rate at the constant LA/St mass ratio of 0.4 and temperature of 55 °C.

contribute to the good dispersion of PCM in the aqueous phase. Therefore, SDS is not much required as a surfactant. The present microencapsulation method of LA could enhance the microencapsulation ratio (ME.R (%)). The microencapsulation ratio could be increased, as compared to that of LA in the previous studies [26,29].

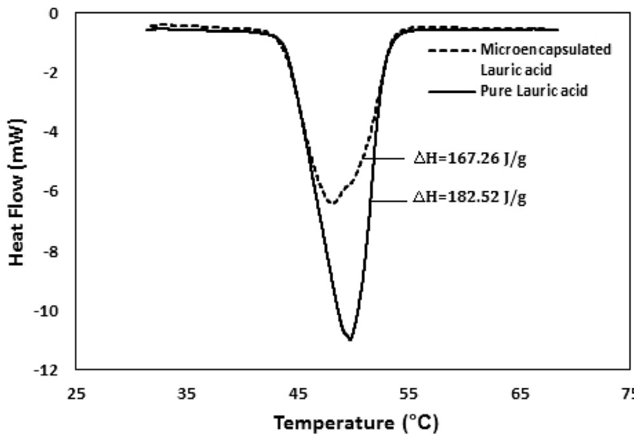


Fig. 5. DSC thermograms of pure lauric acid and the optimal MEPCM.

4. Characterization experiments

4.1. Thermal properties of the optimal microcapsules obtained

Thermal properties such as melting range and latent heat were examined by analyzing DSC thermograms. Fig. 5 shows the DSC thermograms of pure lauric acid and the obtained microencapsulated lauric acid under the optimum conditions. The heat storage capacity of the pure lauric acid was about 182.52 J/g, and the melting point was determined to be about 45 °C. According to Fig. 5, the melting temperature and the latent heat capacity of the optimal microencapsulated phase change material (MEPCM) were detected to be about 43.75 °C and 167.26 J/g, respectively. The optimal microencapsulation ratio (ME.R), calculated by Eq. (1), was 91.64%, which was in a good agreement with the optimal ME.R, as predicted by the quadratic model (92.3%). Fig. 5, also shows that the optimal microcapsules had the same range of phase change as the pure lauric acid.

Thermal stability of the pure lauric acid and the optimal microencapsulated lauric acid was tested by means of TGA and differential thermal (DTA) analyses. The results have been illustrated in Fig. 6. The TGA plot of MEPCM showed a weight loss from 125 °C to 308 °C. The TGA plot of LA showed a one-step weight loss between about 86 °C to 308 °C. Thermal stability of the microencapsulated lauric acid up to 125 °C was much higher than that of the pure LA (86 °C). This that the use of microencapsulation method improved the thermal stability of the LA. The TGA curve of MEPCM exhibited a two-step weight loss behavior. The second step of weight loss took place between 320.46 °C to 459.5 °C, due to the decomposition of polystyrene, as a shell around the LA core.

In the further study, comparison of the weight losing percentage at 150 °C was tested in different time intervals for pure lauric acid and the optimal microencapsulated lauric acid.

5 g of the optimal microcapsules and pure lauric acid were dried at 80 °C until the weight of samples was fixed. Then, the samples put swiftly into an oven at 150 °C. Take the samples out, weight it and recorded every 10 min. The weight losing percentage was calculated as follow:

$$\textit{Percentage of weight loss}\ (\%) = \frac{W_0 - W_n}{W_0} \times 100 \tag{4}$$

where W_n is the weight of samples after curing for n times (n × 10 min) and W_0 is the initial weight of the samples.

Figure 7 shows the percentages of weight loss of the pure lauric acid and optimal microencapsulated LA samples at 150 °C temperature for about 10–50 min. The maximal weight loss percentage for 50 min heating was about 19% for optimal microencapsulated PCM and 39% for pure lauric acid. The rate of weight loss for pure lauric acid was higher than that of microencapsulated LA. It indicates that the polystyrene shell can provide a good protection for core materials.

4.2. FT-IR analysis of optimal microcapsules

In order to prove the presence of components in the microencapsulated lauric acid, Fourier transform infrared (FT-IR) spectra were applied. Fig. 8 depicts the FTIR spectrums of the pure lauric acid, polystyrene and microencapsulated lauric acid. FT-IR spectra of Lauric acid showed the C=O stretching vibration at 1701 cm^{-1}. The stretching vibration is seen at 1294 cm^{-1} which corresponds to the appearance of C—O band. In the spectrums of pure lauric acid and poly(styrene), the symmetric and anti-symmetric CH_2 stretching bands appear at 2850 and 2918 cm^{-1}.

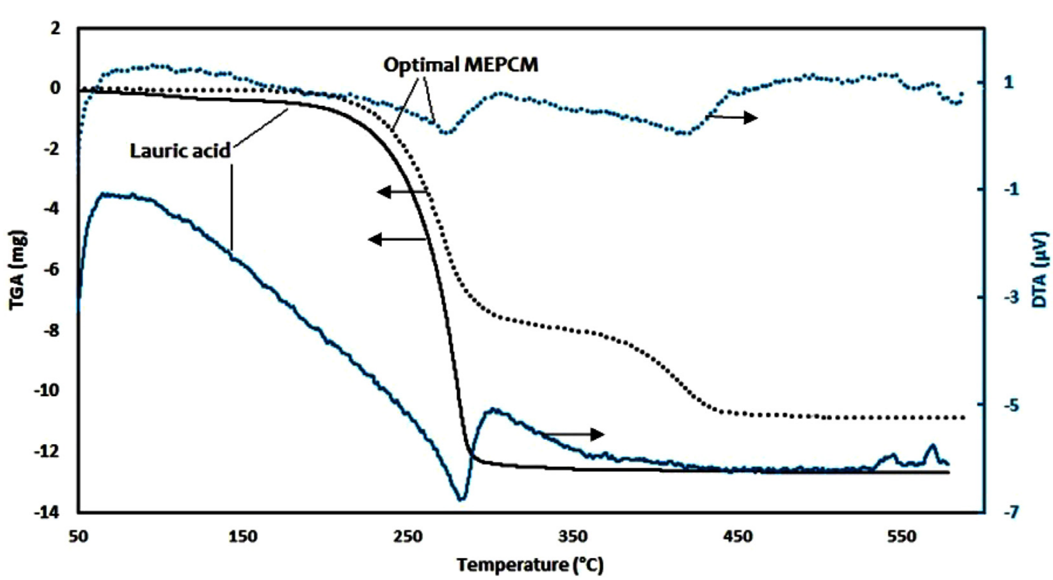


Fig. 6. TGA and DTA of pure lauric acid (LA) and the optimal MEPCM.

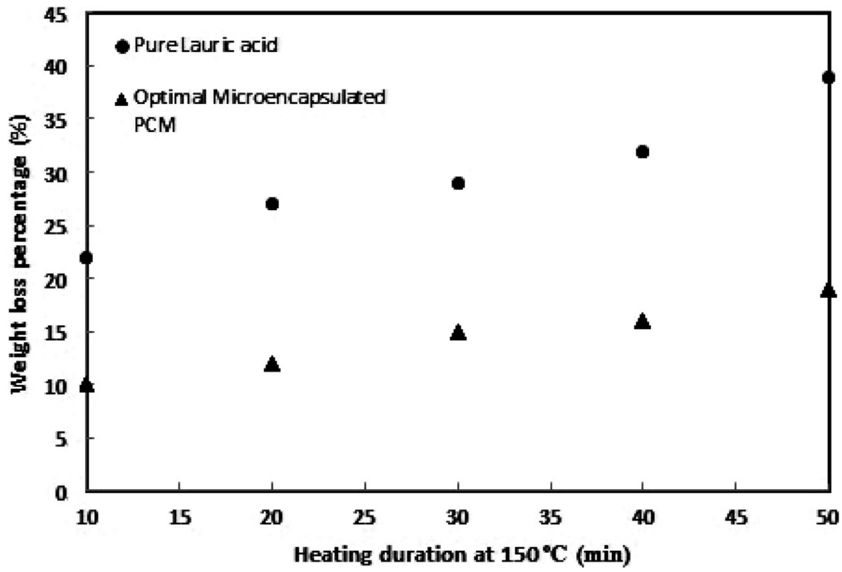


Fig. 7. Comparison of the weight losing percentage under 150 ºC at different time interval for pure lauric acid and optimal microencapsulated PCM.

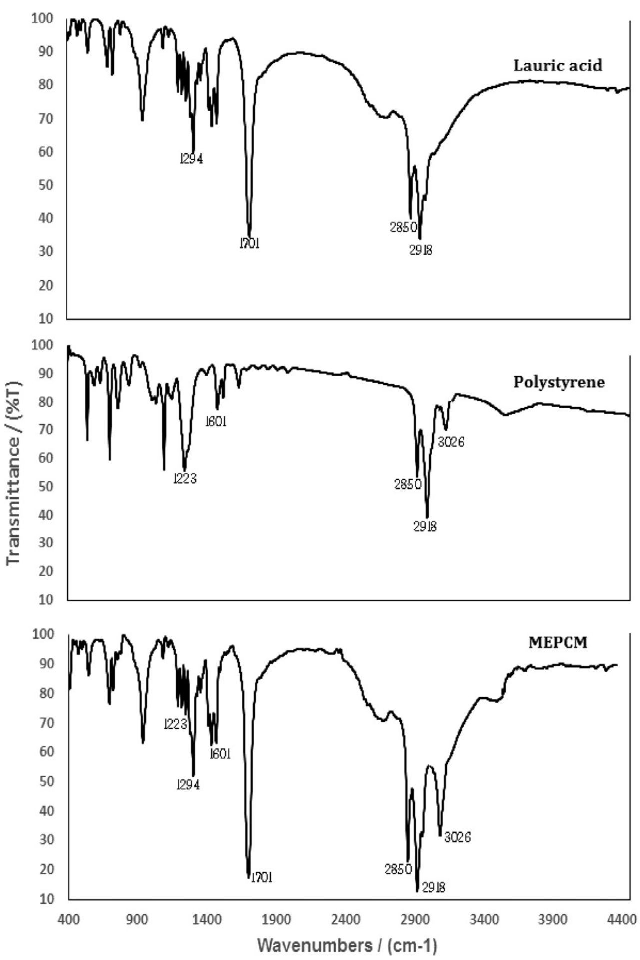


Fig. 8. FT-IR spectra of pure Lauric acid, poly (styrene) and MEPCM.

Poly (styrene) spectrum showed the peak at 3026 cm^{-1} which could be ascribed to ArC-H aromatic stretching vibration of styrene. The other poly (styrene) absorption peak was 1801 cm^{-1}, which corresponded to a weak overtone and C—C out-of-plane (1601 cm^{-1}) [16]. According to Fig. 8, the FT-IR spectra of the MEPCM showed the peaks ascribed to the pure lauric acid (1294, 1701, 2850, 2918 cm^{-1}) and the peaks attributed to the polystyrene (1601, 1223, 2850, 2918, 3026), in the micro particles spectrum.

4.3. Morphology of optimal microcapsules

Fig. 9 shows the micrographs of the obtained microcapsules in the optimum conditions. It could be defined that the microcapsules obtained are spherical, with smooth surfaces, without broken particles or incomplete spheres. Image processing software (Image-J) was used to analyze SEM images and calculate the mean diameter of microcapsules and the coefficient of variation (CV). The results showed that the mean diameter of micro particles was about 1.32 μm, and CV was calculated to be about 0.073. CV (less than one (CV < 1)), which indicates the size distribution has a low-variance and narrow distribution. The mean diameter of micro particles was less than that of the previous studies [29,32].

5. Conclusion

Optimization of the microencapsulation of lauric acid as a renewable and non-toxic PCM and polystyrene as a shell by emulsion polymerization method was investigated, using the response surface method (RSM). The quadratic model was derived for the prediction of the microencapsulation ratio (ME.R). Process optimization was carried out to maximize ME.R. Thus, the optimal values of four factors, including LA/St mass ratio = 0.42, SDS/St mass ratio = 0.01, stirring rate = 1076 rpm and temperature = 55 °C, were calculated by RSM. The maximum ME.R was obtained to be 91.64% in the optimal conditions. The microencapsulation ratio was increased, as compared to that of paraffin in the previous study [32], and also ME.R was enhanced compared with microencapsulation ratio of lauric acid in previous studies. The results showed that the mean diameter of micro particles was about 1.32 μm which was less than that of the previous studies [28,32]. The results of thermal stability show, the rate of weight loss for pure lauric acid was higher than that of microencapsulated LA. These features of microencapsulated of lauric acid cause to improvement of the potential for energy storage and increase the stability of PCM.

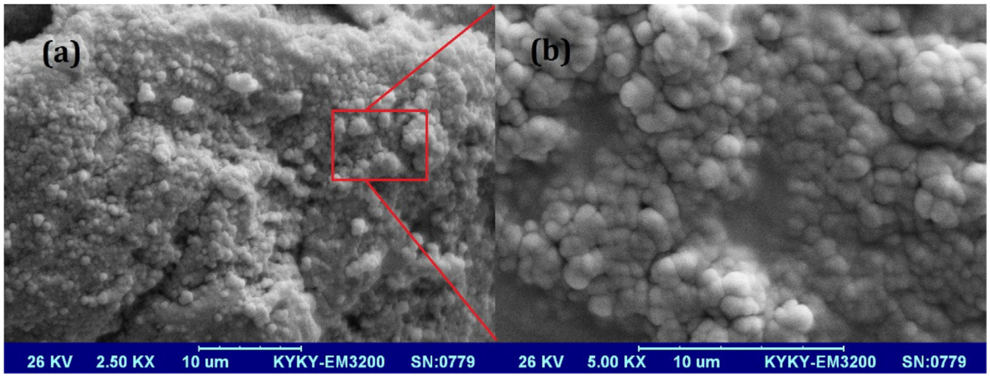


Fig. 9. SEM micrographs of the optimal microcapsules obtained.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.applthermaleng.2017.11.119.

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